

Activity 2

1. Converted to Binary Values (Gender Excluded):

Name	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	1	0	1	0	0	0
Mary	1	0	1	0	1	0
Jim	1	1	0	0	0	0

2. Jaccard Formula

$$Jaccard = \frac{f_{11}}{f_{01} + f_{10} + f_{11}}$$

where:

- f_{11} = number of attributes where both are 1
- f_{01} = number of attributes where first is 0, second is 1
- f_{10} = number of attributes where first is 1, second is 0

3.Jaccaed coefficients:

(Jack, Mary)

Attribute	Jack	Mary	Result
Fever	1	1	$f_{11+} = 1$
Cough	0	0	ignore
Test-1	1	1	$f_{11+} = 1$
Test-2	0	0	ignore
Test-3	0	1	$f_{01+} = 1$
Test-4	0	0	ignore

- $f_{11} = 2$
- $f_{10} = 0$
- $f_{01} = 1$

$$\text{Jaccard Distance} = \frac{1 + 0}{1 + 0 + 2} = \frac{1}{3} \approx 0.33$$

(Jack, Jim)

Attribute	Jack	Jim	Result
Fever	1	1	$f_{11+} = 1$
Cough	0	1	$f_{01+} = 1$
Test-1	1	0	$f_{10+} = 1$
Test-2	0	0	ignore
Test-3	0	0	ignore
Test-4	0	0	ignore

- $f_{11} = 1$
- $f_{10} = 1$
- $f_{01} = 1$

$$\text{Jaccard Distance} = \frac{1 + 1}{1 + 1 + 1} = \frac{2}{3} \approx 0.67$$

(Jim, Mary)

Attribute	Jim	Mary	Result
Fever	1	1	$f_{11}+ = 1$
Cough	1	0	$f_{10}+ = 1$
Test-1	0	1	$f_{01}+ = 1$
Test-2	0	0	ignore
Test-3	0	1	$f_{01}+ = 1$
Test-4	0	0	ignore

- $f_{11} = 1$
- $f_{10} = 1$
- $f_{01} = 2$

$$\text{Jaccard Distance} = \frac{1 + 2}{1 + 2 + 1} = \frac{3}{4} = 0.75$$

Final Answers:

$$\text{Jack \& Mary} = 0.33$$

$$\text{Jack \& Jim} = 0.67$$

$$\text{Jim \& Mary} = 0.75$$